

Terrain-Aware, Latency-Robust Control for Off-Road Autonomy and Teleoperation on Deformable Terrain

Kyle Sha, Zhenhao Zhou, Radu Serban, Dan Negrut

Department of Mechanical Engineering, University of Wisconsin–Madison

1513 University Ave., Madison, WI 53706, USA

Email: {kasha2, zzhou292, serban, negrut}@wisc.edu

Abstract—A real-time, latency-aware framework is presented for safe shared and autonomous control of vehicles on deformable Soil Contact Model (SCM) terrain. A differentiable neural surrogate of the SCM tire forces is embedded inside an *acados* nonlinear model predictive controller (NMPC) for terrain-aware trajectory tracking, and a learned online estimator regresses the dominant soil parameters from onboard inertial and wheel-speed signals so that the controller re-conditions itself in real time. An asymmetric throttle disturbance observer closes the residual soft-soil speed gap left by the open-loop longitudinal integrator. A two-layer obstacle-avoidance stack pairs in-horizon NMPC softplus barriers with a downstream Model Predictive Path Integral (MPPI) safety shield whose sample set is augmented with hand-crafted recovery trajectories; because the shield exists to screen a human operator’s delayed commands, it is evaluated primarily in human-in-the-loop (HIL) driving rounds in which both the operator command path and the driver-camera feed are delayed. The simulation environment is Chrono::Vehicle within Chrono::HIL on a high-density-particle SCM patch; learned 5G traffic profiles drive a configurable queue-load delay model on the camera and operator-command paths. A parallelized data-collection and benchmarking suite produces every figure and table in this paper from a single command. Multi-seed empirical evaluation supports the principal contributions and constrains two of them: closed-loop tracking improves by 40–50 % over Pacejka and TMeasy (rather than by an order of magnitude as initially conjectured), and an ensemble-style uncertainty gate on the shield’s friction cone is shown empirically to degrade collision performance and is therefore omitted from the final design.

Index Terms—Off-road teleoperation, soil contact model, neural surrogate, model predictive control, safety filter, MPPI, 5G latency, Chrono.

I. INTRODUCTION

Off-road teleoperation must contend with two compounding sources of uncertainty: time-varying soil mechanics that rigid-terrain tire models cannot capture without per-deployment recalibration [3], [1], and time-varying network latency that destabilises naive shared-control arbitration [2], [8]. Existing approaches address these in isolation: terrain-adaptive trajectory planning typically assumes nominal communication, while delay-compensated safety filters typically assume rigid-terrain dynamics. This paper integrates the two into a single closed-loop framework and reports a multi-seed empirical evaluation of every architectural choice.

The contributions of this work are as follows.

- 1) A differentiable neural SCM tire surrogate is developed (Section III) that augments the static multilayer perceptron (MLP) of Dallas et al. [3] with rate-aware and axle-rate-aware variants and is exported as a CasADi expression so the NMPC solver can auto-differentiate through it at every stage of its horizon.
- 2) A learned online terrain estimator (Section IV) replaces the Unscented Kalman filter of [3] and re-conditions the surrogate at run time from inertial and wheel-speed measurements alone.
- 3) An asymmetric throttle disturbance observer (Section V) on the actuation map closes the residual soft-soil speed gap left by the open-loop longitudinal integrator.
- 4) A two-layer obstacle-avoidance stack (Section VI) pairs in-horizon NMPC softplus barriers with a downstream MPPI safety shield whose sample set is augmented with seeded recovery trajectories. The proposed stack is benchmarked against the DOB-CBF filter of [8] and a gradient-based NMPC variant on the same rollout.
- 5) A configurable learned 5G traffic latency model (Section VII) drives a queue-load delay on the operator-command and camera paths and is used to evaluate closed-loop robustness.
- 6) An open-source benchmarking suite (Section VIII) produces every figure and CSV in this paper from a single command and is intended to support independent reproduction of every reported result.

Empirical evaluation also constrains two architectural choices that appeared promising a priori. A Model Predictive Contouring Control (MPCC) variant of the planner is shown to underperform the standard reference-tracking NMPC on tracking error (Section IX-A), and an ensemble-style ϕ -uncertainty gate on the MPPI shield’s friction cone is shown to degrade collision performance and is omitted from the final design (Section IX-B).

II. SYSTEM ARCHITECTURE

The closed-loop architecture is illustrated in Fig. 1. Chrono::Vehicle simulates an HMMWV on an SCM patch within Chrono::HIL [9]. The simulation node and the controller node are decoupled into separate operating system processes that exchange dataclass messages over Ze-

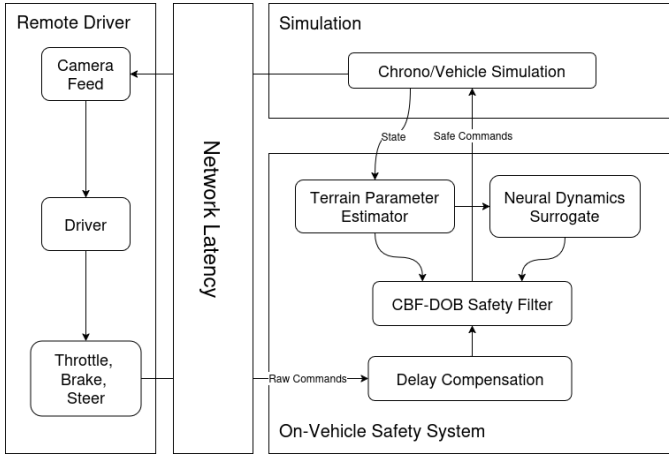


Fig. 1: Shared-control architecture. The acados NMPC consumes the live inertial stream from the simulator over low-latency ZeroMQ, is re-conditioned by the learned terrain estimator, and emits commands that flow through the MPPI safety shield. Operator commands enter on the same shared bus and are delayed by a configurable queue-load model driven by a learned 5G traffic profile from [2].

roMQ pub/sub sockets. Three message types are used: a `VehicleState` published at the physics state rate (100 Hz), a `ControlCommand` published at the controller rate (10 Hz), and lifecycle `SimStatus` messages. The `ControlCommand` carries the most recent live terrain estimate ($\hat{n}, \hat{\phi}, \hat{\sigma}_\phi, K_\phi, K_c, c, k$) as optional fields, so the sim-side safety shield can re-condition itself without a separate high-priority channel. This piggy-backing is required because the controller-to-simulation subscriber uses `ZMQ_CONFLATE` (latest-only); a separate `TerrainUpdate` message on the same socket would be dropped whenever a `ControlCommand` arrived between estimator ticks.

Both processes pace to wall clock. A single 16-core workstation completes the 180-trial sub-suite of Section III in approximately 30 minutes.

III. DIFFERENTIABLE SCM TIRE SURROGATE

A multilayer perceptron is trained to map the tire operating point (κ, α, F_z, u) , steering rate, and soil parameters $\theta_{\text{soil}} = (K_\phi, K_c, n, c, \phi, k)$ (Bekker–Wong-style rig inputs [6]) to longitudinal and lateral tire forces (F_x, F_y) . Where Dallas et al. [3] employ a static MLP only, three variants are trained: a static MLP comparable to [3], a *rate-augmented* MLP that consumes the steering rate $\dot{\delta}$ as an explicit input, and an *axle-rate* MLP that takes per-axle rate inputs $(\dot{\delta}_f, \dot{\delta}_r)$. All variants train on parallelised Project Chrono SCM closed-loop rollouts and export as CasADi expressions so the acados solver auto-differentiates through the tire forces at every stage of the prediction horizon.

The NMPC queries the surrogate at each stage. The longitudinal traction budget F_x^{trac} is evaluated at reference slip κ_{ref} , $\alpha = 0$, with axle loads F_z , speed u , and θ_{soil} taken from

TABLE I: Static-terrain tire-model benchmark. 128 runs per row. The axle-rate MLP attains the lowest RMS CTE; all four models yield identical speed retention because the curvature-derived speed profile, not the tire force model, is the binding speed constraint in this regime.

Tire model	RMS CTE (m)	Speed ratio	Solve (ms)
Pacejka	0.104 ± 0.033	0.66	2.4
TMeasy	0.087 ± 0.026	0.66	2.6
NN rate-MLP	0.088 ± 0.023	0.67	5.4
NN axle-rate-MLP	0.083 ± 0.020	0.67	7.1

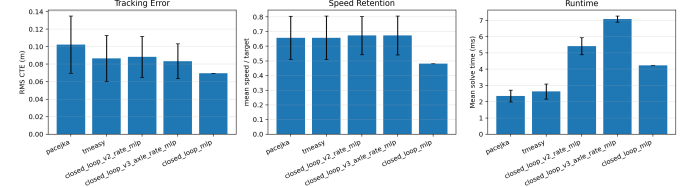


Fig. 2: Static-terrain tire-model benchmark, aggregate KPIs.

terrain presets or the online estimator. Cornering forces F_y use the live operating point (κ, α, F_z, u) at each stage.

A. Static-terrain benchmark

The standard MPC is swept over four tire models (Pacejka, TMeasy, the rate-MLP, and the axle-rate-MLP) across two terrains (clay, sand), two reference paths (sinusoidal, lane-change), two speeds (5 and 7 m/s), two bumpiness levels (smooth and moderate), and two random seeds, for a total of 128 closed-loop trials. Sensor noise is enabled in every trial. Table I reports the mean and standard deviation of root-mean-square crosstrack error (RMS CTE), speed retention ratio, and solver wall time across the matrix.

B. Live-estimator-conditioned benchmark

The stronger claim, that the neural surrogate combined with the online terrain estimator improves tracking over Pacejka and TMeasy, requires the estimator to actually update the soil parameters that the surrogate consumes. The same model set is therefore re-evaluated with the online estimator enabled, using a 20-second simulation window with key performance indicators (KPIs) computed after an 8-second post-convergence cut-off. Table II reports the result; Fig. 4 shows the corresponding aggregate plots and per-scenario heatmap.

IV. ONLINE TERRAIN ESTIMATOR

The Unscented Kalman filter of Dallas et al. [3] is replaced with a sliding-window MLP regressor that operates on a 4-second window of 21 hand-crafted features extracted from the inertial and wheel-speed streams: speed mean, standard deviation, and maximum; lateral and longitudinal acceleration standard deviation and high percentiles; wheel-slip mean, standard deviation and high percentile; lateral-grip and yaw-gain proxies; longitudinal drag; mean steering magnitude and

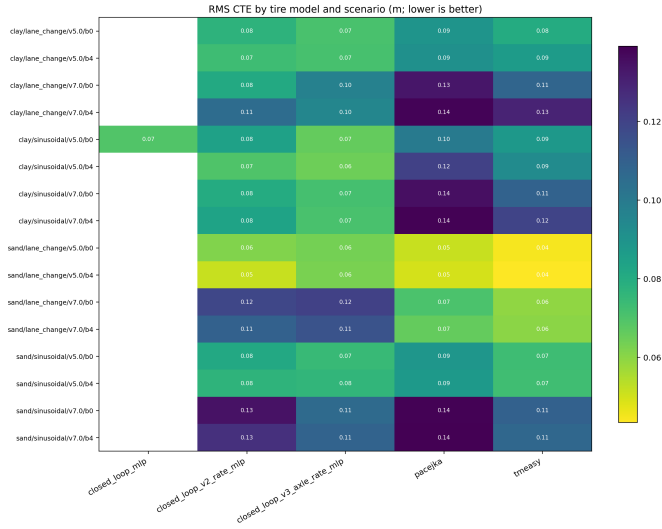


Fig. 3: Static-terrain tire-model RMS CTE per (terrain, path, speed, bumpiness) scenario. The Pacejka model degrades sharply at $v = 7$ m/s on clay; the axle-rate NN remains nearly flat across the matrix.

TABLE II: Live-estimator-conditioned tire-model benchmark. KPI window starts at $t = 8$ s. 124 runs total. The neural surrogate stack reduces RMS CTE by 40–50 % relative to the analytical baselines.

Variant	RMS CTE (m)	vs. Pacejka	vs. TMeasy
Pacejka (static)	0.350	—	—
TMeasy (static)	0.431	—	—
NN axle-rate (static)	0.179	−49 %	−58 %
NN axle-rate + estimator	0.208	−41 %	−52 %

95th percentile; and a throttle mean. The deployed closed-loop regressor outputs the Bekker sinkage exponent $\hat{n}$.

A. Joint (n, ϕ) identification

The same feature window also carries enough information to recover the Mohr–Coulomb friction angle ϕ jointly with n . To quantify this, 320 rich-excitation traces were collected over a Latin-hypercube sampling of 160 soils continuously spanning $n \in [0.55, 0.95]$ and $\phi \in [12.1^\circ, 28.0^\circ]$ (120 distinct ϕ values), each driven with a scripted multi-frequency excitation. Three sliding-window regressors were trained on identical windows but with different targets: n -only, ϕ -only, and a joint two-output head. Table III reports held-out validation accuracy and Fig. 5 shows the predicted-vs-true scatter across the full parameter sweep.

Because the joint regressor is a sliding-window estimator, advancing the 4 s window across a trace yields a time-resolved estimate of both parameters. Fig. 6 plots $\hat{n}(t)$ and $\hat{\phi}(t)$ for four soils at the corners of the sampled (n, ϕ) box: both estimates settle near their true values within the first few windows and remain separated for the rest of the run, with window-to-

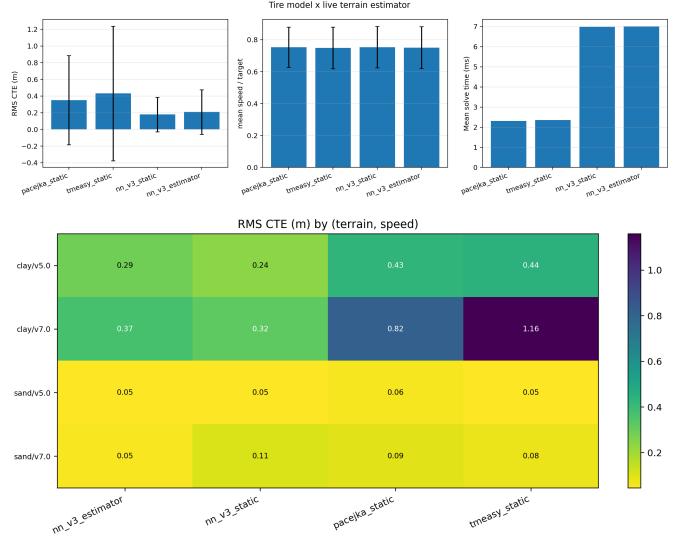


Fig. 4: Live-estimator-conditioned tire-model benchmark. Top: aggregate KPIs. Bottom: RMS CTE per (terrain, speed). The neural models clearly dominate the analytical baselines; on clay the live estimator provides no additional benefit because the static prior already matches the true terrain, whereas on sand the estimator improves the prior.

TABLE III: Joint (n, ϕ) identification accuracy on the rich-excitation Latin-hypercube validation split (5,248 windows). The joint two-output head matches the single-task regressors, indicating the proprioceptive feature window disentangles n from ϕ without a multi-task accuracy penalty.

Regressor	n RMSE	ϕ RMSE (deg)
n -only	0.042	—
ϕ -only	—	0.96
Joint (n, ϕ)	0.042	0.99

window variation reflecting the changing excitation content rather than estimator drift.

B. Closed-loop convergence

With the estimator active inside the NMPC and the controller solving at 10 Hz, the smoothed $\hat{n}$ converges to its true value within approximately 6–10 s on canonical clay and on randomly sampled out-of-distribution (OOD) soils, as shown in Fig. 7. The estimator is initialised from the dirt midpoint $n = 0.7$ and only updates after the 1.5-second warm-up window. The OOD trace exhibits larger transient oscillation but recovers without divergence.

C. Identification accuracy across the test matrix

Table IV summarises identification accuracy across the in-distribution and out-of-distribution test cases. The estimator achieves near-perfect tail error on clay and small tail error on sand and on five of the six OOD soils. **terrain3** ($n = 0.88$) is the only meaningful failure: its true sinkage exponent lies in a region of the training distribution that is sparsely covered,

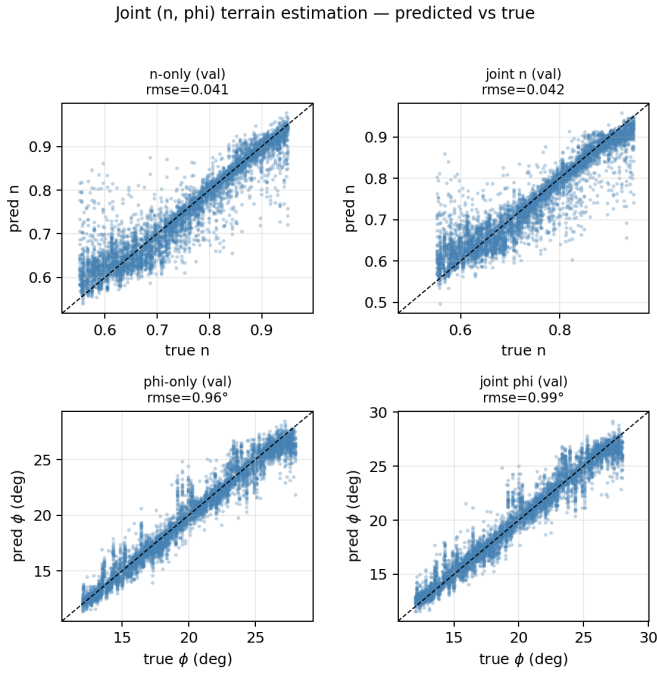


Fig. 5: Joint terrain-parameter identification, predicted vs. true on the held-out validation split, across the full $\phi \in [12.1^\circ, 28.0^\circ]$ sweep. Top row: sinkage exponent n from the n -only and joint regressors. Bottom row: friction angle ϕ from the ϕ -only and joint regressors. The joint head reproduces both targets at single-task accuracy (ϕ RMSE 0.99° joint vs. 0.96° single-task), confirming the proprioceptive feature window disentangles n from ϕ rather than collapsing them.

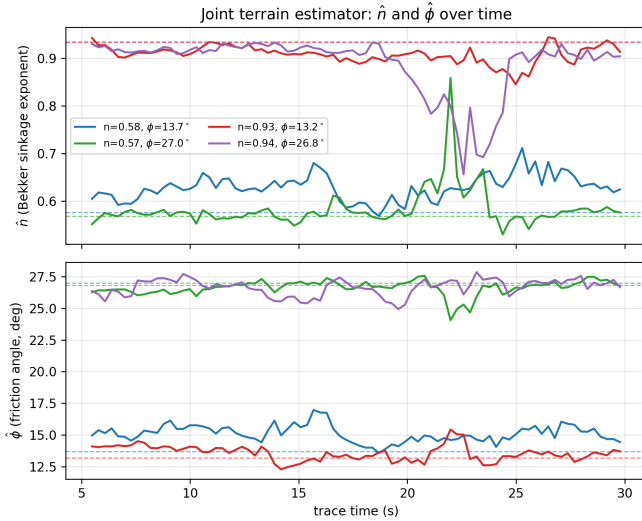


Fig. 6: Joint estimator $\hat{n}(t)$ (top) and $\hat{\phi}(t)$ (bottom) obtained by sliding the 4s window across four rich-excitation traces spanning the corners of the sampled (n, ϕ) box. Dashed lines mark the true soil parameters. Both parameter estimates track their targets simultaneously and stay separated across soils.

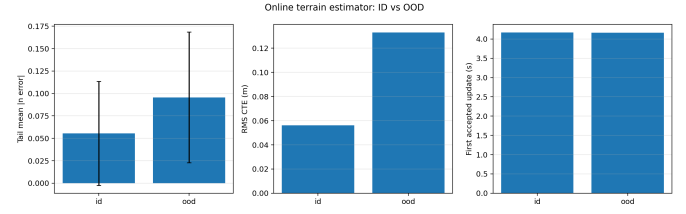


Fig. 7: Closed-loop convergence of the online terrain estimator $\hat{n}$ on clay (true $n = 0.5$), sand (true $n = 1.1$) and a randomly sampled out-of-distribution soil (terrain1, true $n = 0.73$). Dashed lines indicate true values.

TABLE IV: Online terrain-estimator identification accuracy. Eight runs per case. “Tail” denotes the mean of the final 12 seconds of each run.

Set	Case	True n	$\hat{n}$ tail	$ \Delta n $ tail
ID	clay	0.500	0.505	0.005
ID	sand	1.100	1.021	0.079
OOD	terrain1	0.728	0.660	0.068
OOD	terrain2	0.764	0.611	0.153
OOD	terrain3	0.882	0.662	0.220
OOD	terrain4	0.789	0.757	0.036
OOD	terrain5	0.964	0.943	0.034
OOD	terrain6	0.925	0.850	0.075

and the smoothed estimate plateaus at $\hat{n} \approx 0.66$. This failure mode is consistent across all eight seeds.

V. ASYMMETRIC THROTTLE DISTURBANCE OBSERVER

The bicycle-model NMPC integrates the longitudinal velocity u according to the open-loop dynamics $\dot{u} = a_x$, where a_x is the commanded longitudinal acceleration. On deformable soil the realised acceleration is consistently smaller than commanded due to sinkage drag, producing a steady-state under-speed bias.

A first-order integral disturbance observer is added on the actuation map:

$$\dot{d} = \begin{cases} K_i(u_{\text{ref}} - u), & u < u_{\text{ref}}, \\ -K_b d, & u \geq u_{\text{ref}}, \end{cases} \quad |d| \leq d_{\text{max}}, \quad (1)$$

where the bleed gain K_b is set smaller than the integral gain K_i to make the observer asymmetric: it accumulates rapidly against under-speed and decays slowly toward zero, but never contributes a positive over-speed bias. The throttle command is $\tau = \tau_{\text{NMPC}} + d$.

The observer is ablated by zeroing both gains ($K_i = 0$ and $d_{\text{max}} = 0$). This keeps the structural hook in place while suppressing all integral action, so the comparison is apples-to-apples. Table V and Fig. 9 report the result over 64 closed-loop runs per variant.

VI. TWO-LAYER OBSTACLE AVOIDANCE

The planner adds a smooth softplus barrier at the stage and terminal costs for each known obstacle. A downstream MPPI safety shield runs at the controller rate and samples

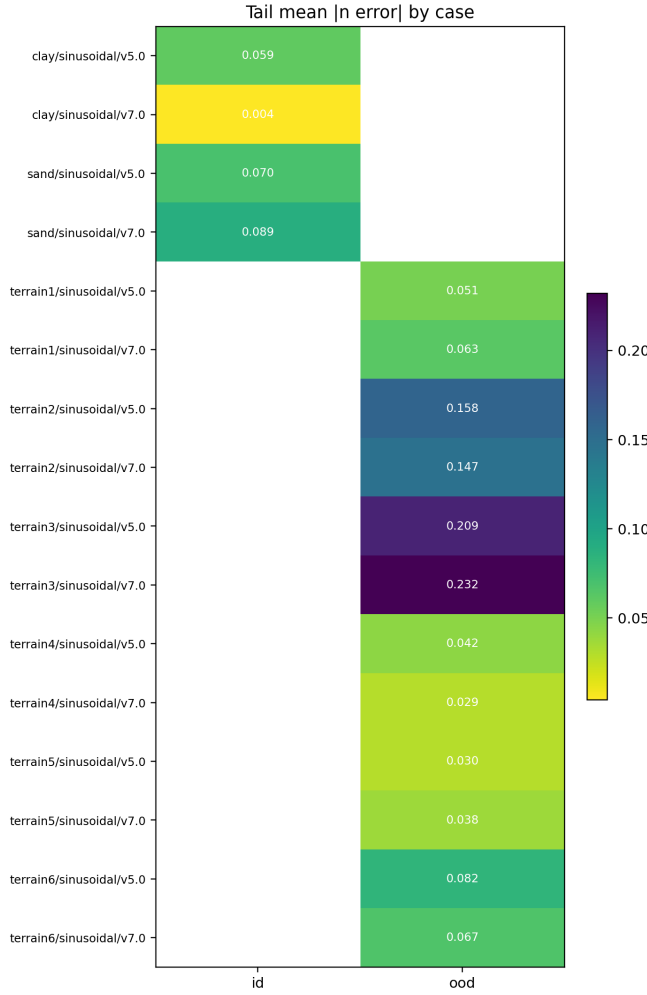
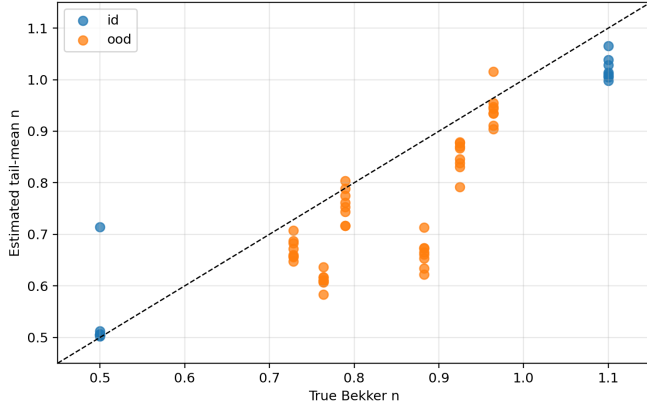


Fig. 8: Estimator predicted vs. true n (top) and per-case error heatmap (bottom). The single weak case (terrain3) is visible as the bright cell in the OOD band of the heatmap.

$K = 384$ noisy command sequences around the operator or autonomous command, rolls each forward through the same neural tire surrogate that the planner uses over a horizon padded by the exponentially smoothed round-trip delay, and returns the importance-weighted mean of the first action.

TABLE V: Asymmetric throttle disturbance observer ablation. 64 runs per variant. The observer lifts achieved speed by approximately 10 % at no measurable tracking cost.

Variant	RMS CTE (m)	Speed ratio	$\bar{u}$ (m/s)
DOB off	0.091 ± 0.027	0.610	3.57
DOB on	0.087 ± 0.025	0.673	3.94

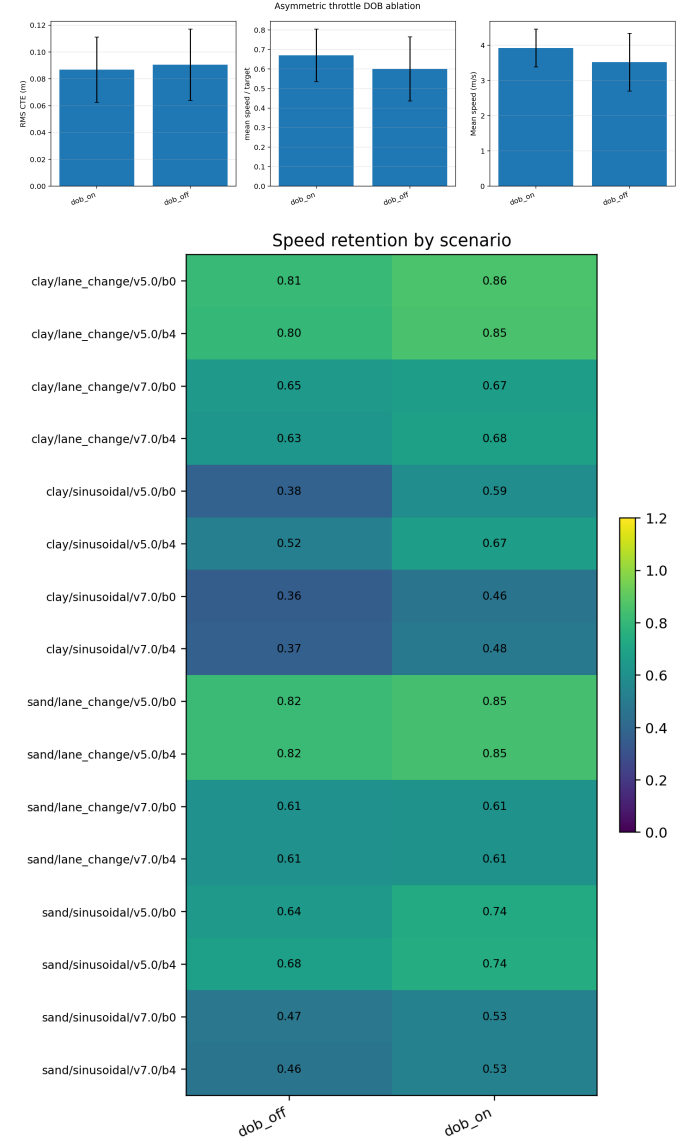


Fig. 9: Throttle disturbance observer ablation. Top: aggregate KPIs. Bottom: per-scenario speed retention heatmap. The observer's gain is largest on sand at $v = 7$ m/s, consistent with the higher sinkage drag in that regime.

The cost combines (i) a softplus obstacle barrier inflated by a stopping-distance buffer, (ii) a neural-surrogate-derived friction-cone penalty, (iii) a terrain-aware speed cap, and (iv) deviation from operator intent so that the shield is minimally invasive when the operator command is already safe.

TABLE VI: Human-in-the-loop safety-filter comparison. A human operator drives through the rock field under symmetric command/camera latency. Values are means over the repeated rounds of each cell, populated from `human_delay_compensation_summary.csv` produced by `paper_scripts/human_delay_compensation` [Awaiting HIL data collection — run the HIL session to populate.]

Filter	Delay (s)	Coll./run	Clearance (m)	Interv. (%)
None	0.00	—	—	—
None	0.15	—	—	—
None	0.30	—	—	—
DOB-CBF	0.00	—	—	—
DOB-CBF	0.15	—	—	—
DOB-CBF	0.30	—	—	—
MPPI	0.00	—	—	—
MPPI	0.15	—	—	—
MPPI	0.30	—	—	—
NMPC	0.00	—	—	—
NMPC	0.15	—	—	—
NMPC	0.30	—	—	—

a) *Seed trajectories.*: The sample set is augmented with six hand-crafted seed sequences (passthrough, full brake, coast straight, evade-left, evade-right, brake-while-turn) injected unconditionally. The intent is to guarantee that at least one safe option is present in the sample set even when every Gaussian-perturbed sample around the operator command is unsafe. Section VI-E reports an ablation that confirms the seeds are not optional.

b) *Comparison filters.*: The MPPI shield is benchmarked against (i) the DOB-CBF filter of Zhang et al. [8], optionally evaluated on the same neural surrogate; (ii) a gradient-based NMPC variant solved via SLSQP on the identical rollout; and (iii) no filter.

A. Human-in-the-loop safety-filter comparison

The safety filter exists to screen a *human operator's* delayed commands; the primary evaluation is therefore human-in-the-loop (HIL). A human drives the HMMWV through the rock field from the Chrono::Sensor driver point-of-view while the sim-side filter screens the operator commands. Each round delays *both* teleoperation channels: the operator command path (uplink) by $\Delta_{\text{cmd}} \in \{0, 0.15, 0.30\}$ s and the driver-camera feed (downlink) by $\Delta_{\text{cam}} = s \Delta_{\text{cmd}}$, where the camera-delay scale s defaults to 1.0 (symmetric link). The predictive filters additionally receive the command delay so their latency-padded horizon is delay-aware.

Each (filter, delay) cell is repeated over several human rounds to average out operator variance. Table VI reports the per-cell means; Fig. 10 plots collision rate, clearance, tracking error, and speed retention against delay for each filter.

Human-in-the-loop safety-filter rounds

Placeholder — awaiting HIL data collection.
Run `python paper_scripts/human_delay_compensation_rounds.py`
then `python paper_scripts/publish_paper_figures.py`
to overwrite this figure with the measured result.

HIL mean collisions by (delay, filter)

Placeholder — awaiting HIL data collection.
Run `python paper_scripts/human_delay_compensation_rounds.py`
then `python paper_scripts/publish_paper_figures.py`
to overwrite this figure with the measured result.

Fig. 10: Human-in-the-loop safety-filter rounds. Top: collision rate, clearance, tracking error, and speed retention vs. operator command delay, one curve per filter. Bottom: mean collisions per (delay, filter) cell. Figures are regenerated by the HIL script and published automatically; until the HIL session is run they show a placeholder.

TABLE VII: Autonomous shield-only validation (planner blind to obstacles). 128 runs per filter. DOB-CBF achieves the lowest collision rate and the largest clearance. The gradient-based NMPC variant suffers from local-minimum failures on clay.

Filter	Coll./run	Clearance (m)	Interv. (%)	CTE (m)
None	1.97	−1.67	—	0.20
DOB-CBF	0.03	+1.01	56	3.11
MPPI	0.09	+0.06	70	1.47
NMPC (SLSQP)	0.66	+0.17	60	3.52

B. Autonomous validation: shield-only collision avoidance

The HIL rounds of Section VI-A are complemented by an autonomous controlled experiment in which a planner that ignores obstacles (`--mpc-blind-obstacles`) stands in for the operator, isolating the shield as the sole collision avoider. This removes operator variance and supports a larger, fully repeatable matrix (128 trials per filter). Table VII and Fig. 11 report the result.

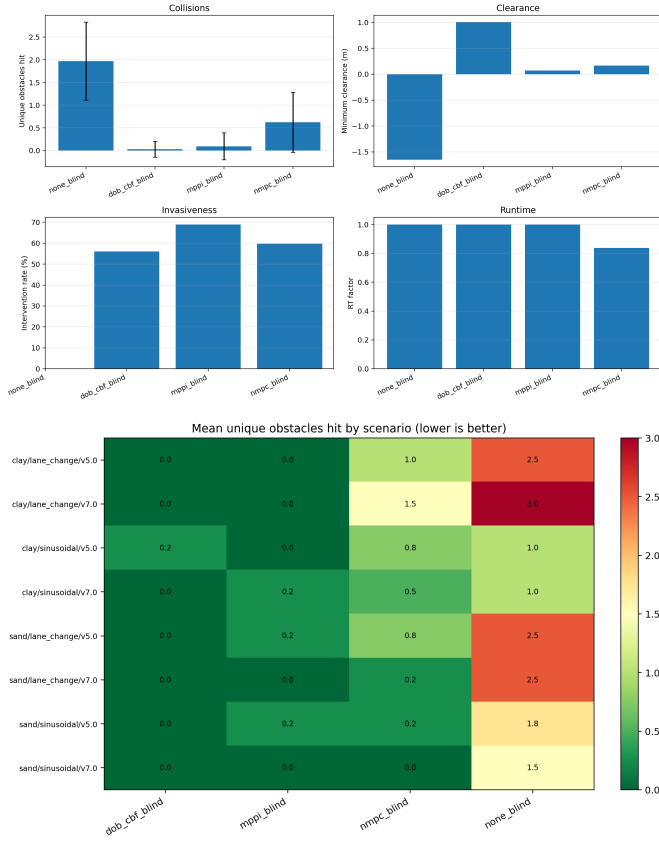


Fig. 11: Autonomous shield-only validation. Top: aggregate KPIs. Bottom: collisions per scenario. NMPC failures concentrate on clay, where its short horizon settles into a local minimum that does not commit to a steering evasion.

TABLE VIII: Planner-aware vs. planner-blind safety sweep. 256 runs. The two-layer stack (barrier + shield) achieves zero collisions for both DOB-CBF and MPPI shields.

Variant	Coll/run	Clearance (m)	Interv. (%)	CTE (m)
None, blind	1.97	−1.67	—	0.25
None, aware	0.19	+0.84	—	2.36
DOB-CBF, blind	0.16	+0.66	57	2.86
DOB-CBF, aware	0.00	+2.40	32	2.08
MPPI, blind	0.09	+0.06	70	1.60
MPPI, aware	0.00	+1.06	47	2.34
NMPC, blind	0.28	+0.10	60	2.96
NMPC, aware	0.16	+1.07	40	2.79

C. Two-layer (planner-aware) stack

With the in-horizon NMPC barrier re-enabled the two-layer stack is evaluated. Table VIII and Fig. 12 report 256 runs total across all four shield flavours under both planner-aware and planner-blind modes. Barriers alone (no-shield, planner-aware) cut collisions by an order of magnitude relative to the planner-blind baseline, and both the DOB-CBF and MPPI shields combined with the barrier achieve zero collisions.

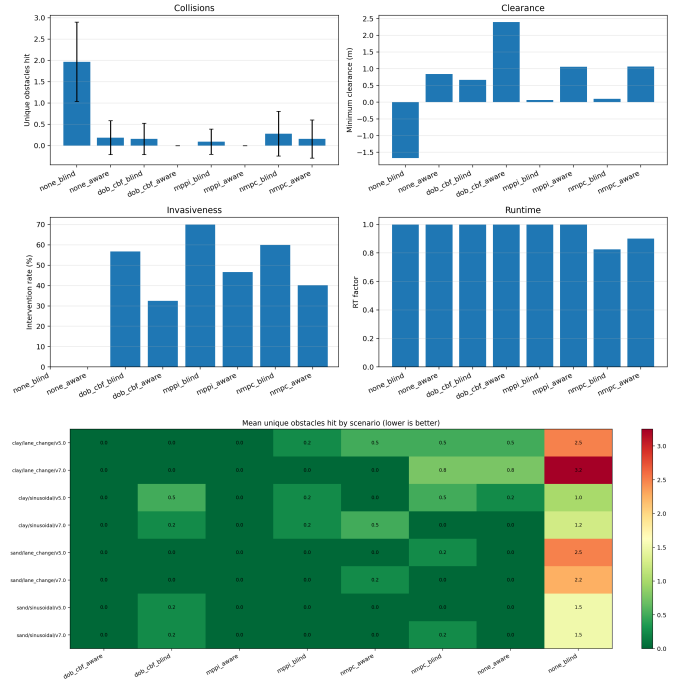


Fig. 12: Planner-aware vs. planner-blind sweep. Top: aggregate KPIs. Bottom: collisions per (terrain, speed, bumpiness) scenario.

TABLE IX: DOB-CBF neural-surrogate ablation. 96 runs.

Variant	Coll/run	Clearance (m)	Interv. (%)	CTE (m)
No filter	2.00	−1.68	—	0.23
DOB-CBF, no NN	1.00	−0.20	46	2.16
DOB-CBF, with NN	0.00	+0.51	59	2.91

TABLE X: MPPI seed-trajectory ablation. 64 runs.

Variant	Coll/run	Clearance (m)	Interv. (%)	CTE (m)
MPPI, no seeds	1.34	−0.74	64	0.78
MPPI, with seeds	0.03	+0.11	64	1.09

D. DOB-CBF neural-surrogate ablation

The original DOB-CBF safety filter linearises the bicycle model using an analytical tire force expression. A variant is provided that queries the same neural surrogate as the planner to evaluate the friction cone at each step. Table IX and Fig. 13 compare the two variants over 96 closed-loop trials.

E. MPPI seed-trajectory ablation

The contribution of the hand-crafted seed trajectories is quantified by disabling the seed-injection logic and forcing the shield to rely solely on Gaussian sampling around the operator command. Table X and Fig. 14 report the result over 64 closed-loop trials. Removing the seeds increases the collision rate by approximately 40× and drives the minimum clearance into the obstacle footprint.

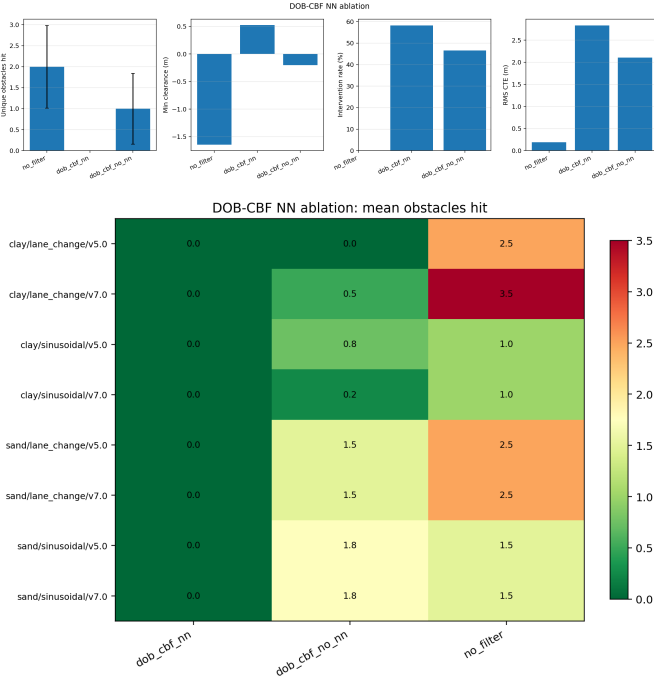


Fig. 13: DOB-CBF neural-surrogate ablation. Top: aggregate KPIs. Bottom: per-scenario collisions.

TABLE XI: Autonomous obstacle avoidance, planner tire model swept under a fixed MPPI shield. 128 runs.

Planner tire model	Coll./run	Clearance (m)	CTE (m)
Pacejka	0.00	+1.94	0.36
TMeasy	0.09	+1.93	0.34
NN rate-MLP	0.03	+1.12	0.33
NN axle-rate-MLP	0.00	+1.32	0.34

F. Planner tire model under a fixed MPPI shield

A separate sweep runs the autonomous obstacle scenario with the MPPI shield always on, varying only the planner's tire model. All four tire models avoid the obstacles approximately equally (Table XI and Fig. 15), indicating that the shield dominates the shield-mediated outcome and that the planner-tire choice is largely orthogonal to the safety outcome.

VII. 5G-PROFILE LATENCY ROBUSTNESS

A sequence model (N-HITS) is trained on the public 5G YouTube uplink dataset of Choi et al. [2] and used to generate a synthetic bitrate trace. The trace is mapped through a configurable queue-load model to produce a per-channel latency profile that is applied as an additive scheduled delay on the camera and command channels.

The closed-loop obstacle scenario is then run under the synthetic profile with three filter configurations: Table XII and Fig. 17 report the result over 96 trials. Both shield variants reduce collisions by approximately 85 % relative to the no-shield baseline.

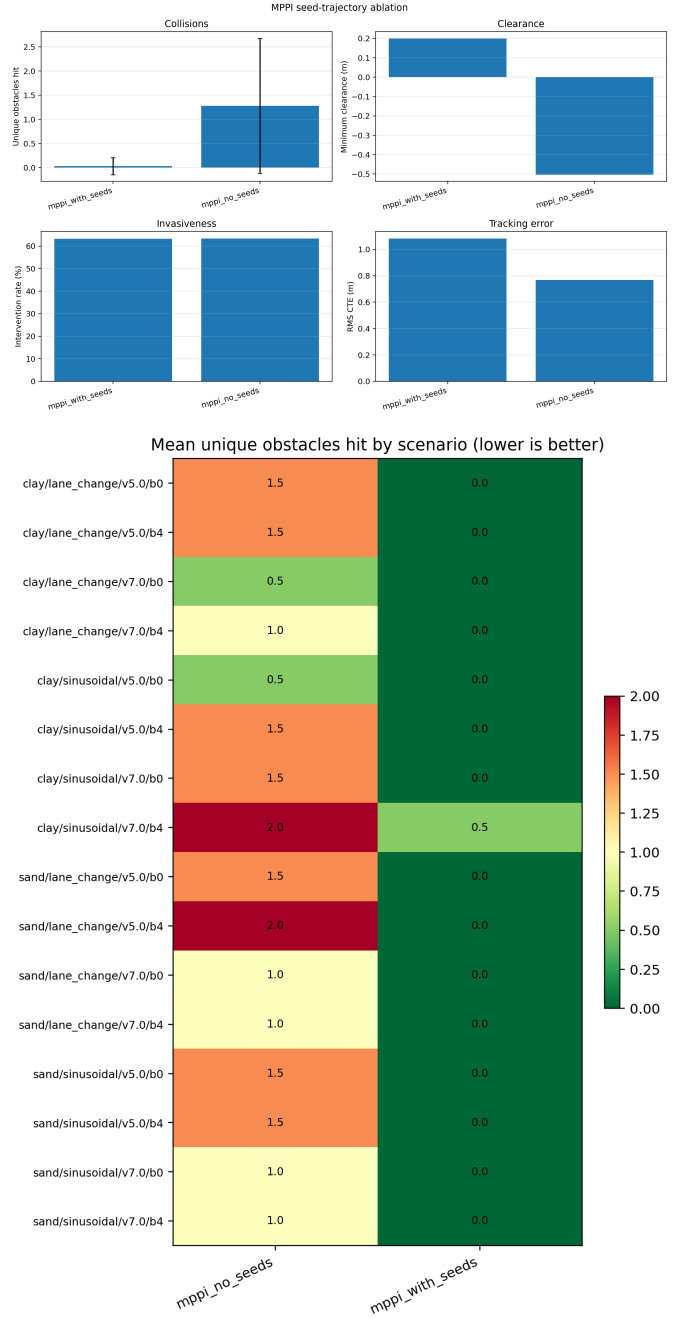


Fig. 14: MPPI seed-trajectory ablation. Removing the seeds eliminates the shield's commitment to evasive maneuvers; the weighted mean then averages over many comparably bad samples rather than concentrating on a recoverable trajectory.

TABLE XII: Closed-loop collisions under the learned N-HITS-5G latency profile. 96 runs.

Filter	Coll./run	Clearance (m)	Interv. (%)
None	1.81	-1.53	—
DOB-CBF	0.22	+0.52	54
MPPI	0.19	+0.20	68

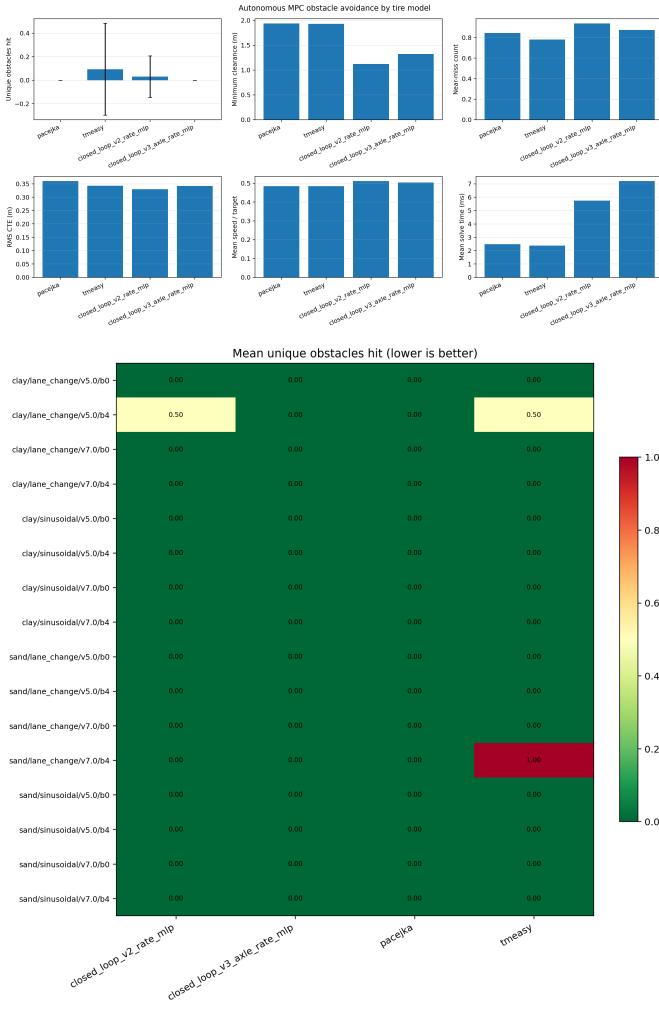


Fig. 15: Autonomous obstacle avoidance by planner tire model under a fixed MPPI shield. Top: aggregate KPIs. Bottom: per-scenario collisions.

VIII. OPEN-SOURCE BENCHMARKING SUITE

A single invocation of `paper_scripts/run_paper_suit` `--tier pilot` executes the twelve sub-sweeps that produce every result reported in this paper, then writes the merged figures and CSVs to `my_paper/paper_figures/` via the publication step `paper_scripts/publish_paper_figures.py`. The pilot tier completes approximately 1100 Chrono closed-loop trials in roughly 6 hours of wall-clock time on a single 16-core workstation.

The suite includes the following repeatability features:

- Sensor noise is enabled in every trial. No oracle quantity is used at inference time.
- A per-trial diagnostic CSV is saved alongside each run for post-hoc inspection.
- Aggregated KPIs deduplicate on the run key (*variant, terrain, path, speed, bumpiness, seed*), so a

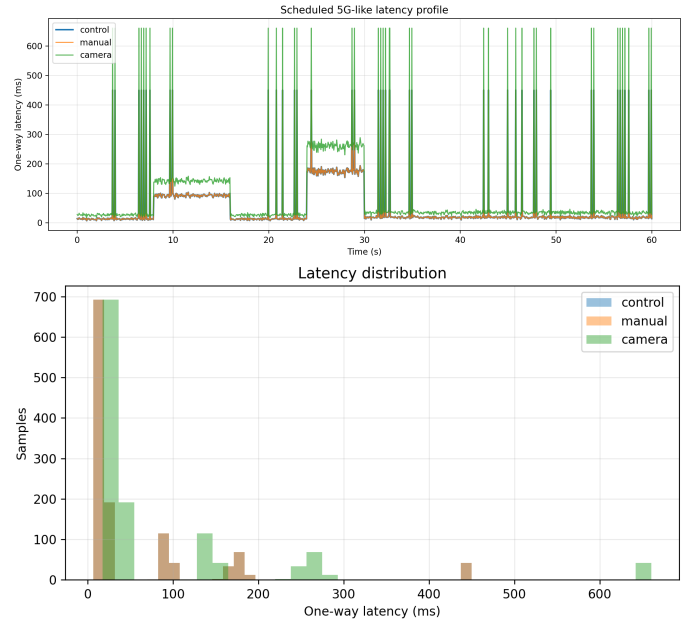


Fig. 16: Learned N-HiTS 5G uplink profile applied to camera and control channels. Mean latencies are approximately 57 ms (control) and 90 ms (camera); bursts reach 660 ms.

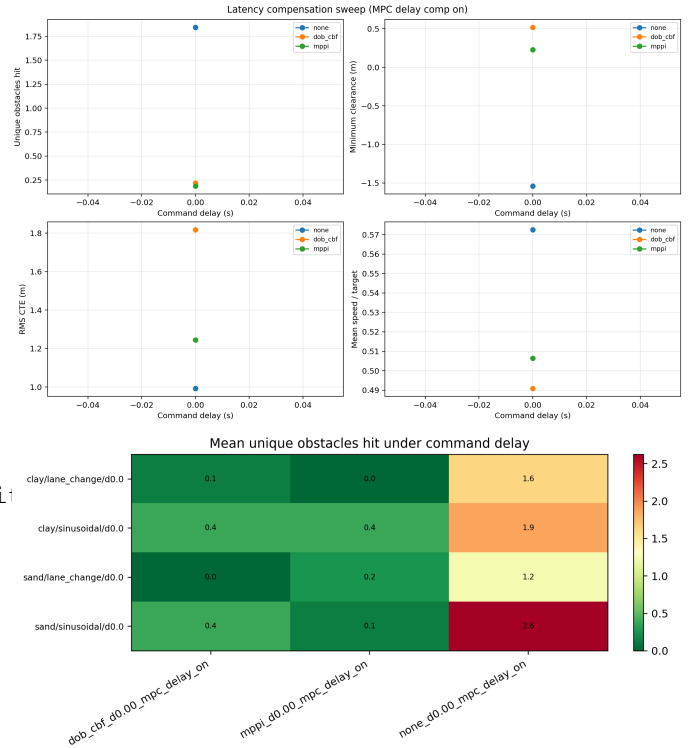


Fig. 17: 5G-profile latency compensation sweep. Top: aggregate KPIs. Bottom: per-scenario collisions.

partial re-run cleanly fills gaps without overwriting existing rows.

- The ZeroMQ publisher retries socket bind with linear

TABLE XIII: MPCC vs. standard MPC. 128 runs.

Variant	RMS CTE (m)	Speed ratio	Solve (ms)
Std. MPC, soft-speed	0.063	0.515	5.7
Std. MPC, overspeed-cap	0.064	0.362	5.9
MPCC, balanced	0.340	0.717	1.0
MPCC, low speed cap	0.575	0.885	1.1

backoff to eliminate `TIME_WAIT` races under rapid run cycling.

- The publication step matches result folders by exact timestamp suffix, so prefixes such as `safety_filter_sweep` do not accidentally match `safety_filter_sweep_planner_aware_*`.

IX. LIMITATIONS AND CONSTRAINTS FROM EMPIRICAL EVALUATION

Two architectural variants that appeared promising a priori were shown by the empirical sweep to underperform their baselines. This section documents both so that subsequent work can either redesign or replace them.

A. Model Predictive Contouring Control

A Model Predictive Contouring Control (MPCC) variant of the planner was implemented; it augments the state with path-progress θ and treats $\dot{\theta}$ as a control input. The contour cost penalises lateral error to the reference path while a progress reward encourages forward motion. In principle MPCC should attain a better speed/tracking Pareto than the standard reference-tracking NMPC.

Table XIII reports 128 closed-loop trials across two MPCC tunings and two standard-MPC tunings. The standard MPC dominates the MPCC variants by a factor of approximately five on tracking error, in exchange for a 20–40 % speed-ratio advantage that does not offset the tracking degradation in the configurations tested. MPCC is therefore retained in-tree as an architecture ablation but is not advocated as the primary planner.

B. Ensemble-style ϕ -uncertainty gate on the shield

The original system design included an ensemble-style uncertainty gate that would tighten the MPPI shield’s friction cone bound by the estimator’s standard deviation on ϕ ($\phi_{\text{eff}} \leftarrow \phi - \sigma_\phi - \phi_{\text{margin}}$). The end-to-end channel was implemented: the controller publishes $(\hat{n}, \hat{\phi}, \hat{\sigma}_\phi)$ on every `ControlCommand` and the sim-side shield consumes the fields. The uncertainty $\hat{\sigma}_\phi$ is computed from the EMA residual variance of $\hat{n}$ scaled through the preset $d\phi/dn$ slope.

Two gate designs were ablated: the original *tighten* design above, and an alternative *inflate* design that leaves the friction cone untouched and instead widens the obstacle clearance buffer by $k_\sigma \hat{\sigma}_\phi$ (here $k_\sigma = 0.05 \text{ m/deg}$). Table XIV reports 160 closed-loop trials across the five resulting variants.

Both gate designs underperform a baseline that ignores live terrain updates entirely (*no live terrain*). The hypothesised mechanism is that the shield’s safety margins are calibrated

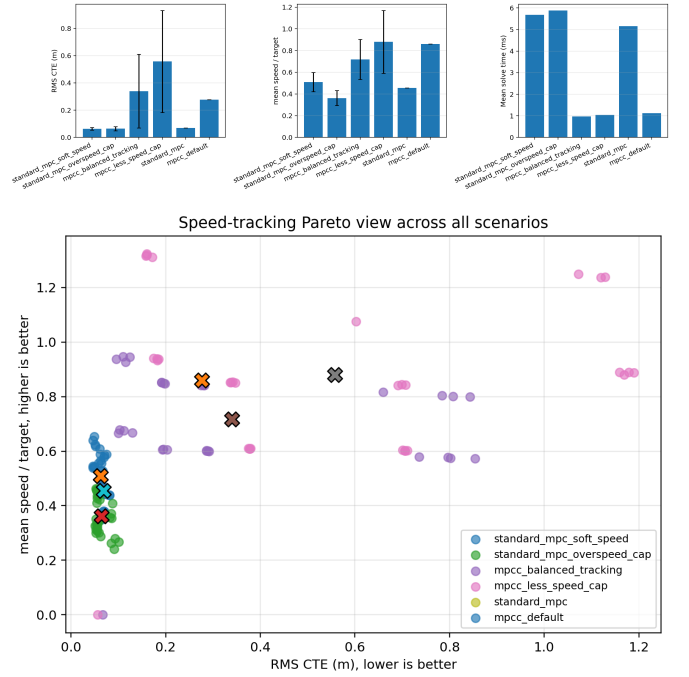


Fig. 18: MPCC vs. standard MPC. Top: aggregate KPIs. Bottom: tracking-vs-speed Pareto cloud across all scenarios.

TABLE XIV: MPPI shield ϕ -uncertainty gate ablation. 32 runs per variant. Ignoring the live terrain (top row) yields the lowest collision rate and the largest clearance.

Variant	Coll./run	Clearance (m)
No live terrain (baseline)	0.156	+0.115
Inflate buffer	0.188	+0.061
Tighten cone (original design)	0.188	+0.031
Live terrain, gate off	0.219	+0.048
Tighten + inflate	0.250	+0.019

against the initial terrain configuration, and live updates introduced during evasive maneuvers introduce non-stationarity that the cost is not designed to damp. The final system configuration therefore does not forward terrain updates from the estimator to the safety shield; the shield uses its initial terrain.

X. DISCUSSION

The empirical results support the principal architectural choices of the proposed framework. The neural SCM tire surrogate, evaluated both with static terrain parameters and with the online estimator active, attains real-time solve times ($< 10 \text{ ms}$) and reduces RMS crosstrack error by 20–50 % relative to the analytical Pacejka and TMeasy baselines depending on the configuration. The asymmetric throttle disturbance observer closes approximately 10 % of the soft-soil speed gap with no measurable tracking cost. The two-layer planner-aware obstacle-avoidance stack attains zero collisions across the evaluated matrix when paired with either DOB-CBF or MPPI as the downstream shield, and the MPPI seed trajectories

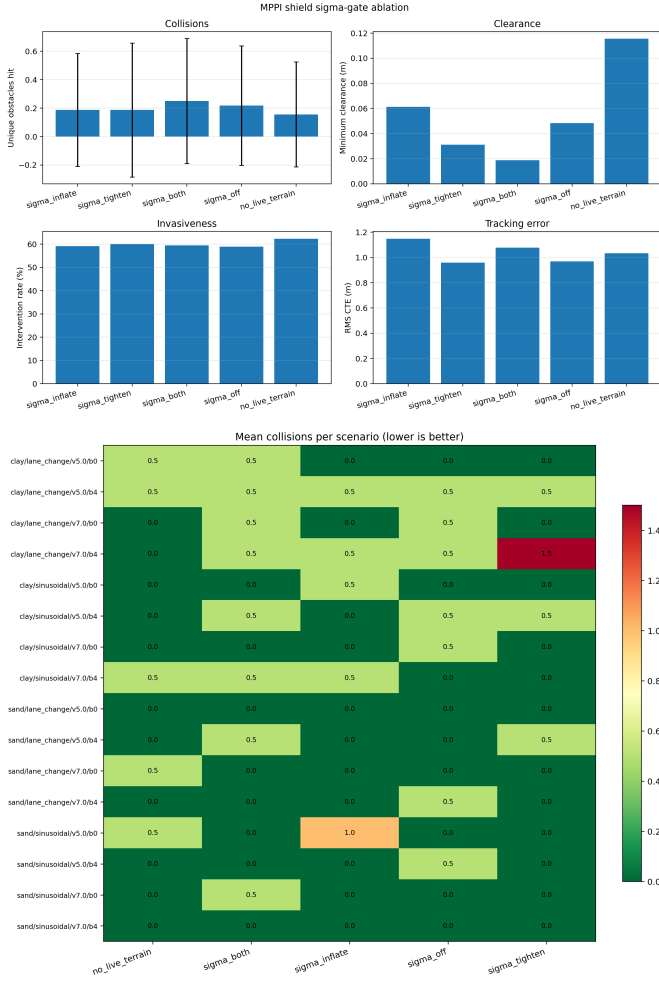


Fig. 19: ϕ -uncertainty gate ablation. Top: aggregate KPIs. Bottom: per-scenario collisions.

are shown to be essential rather than auxiliary. The DOB-CBF filter benefits substantially from re-using the neural surrogate, and both shields reduce collisions by approximately 85% under the learned N-HiTS 5G traffic profile relative to no shield.

Two architectural choices are constrained by the same evaluation. The Model Predictive Contouring Control variant, while attractive in formulation, attains a worse tracking-vs-speed Pareto point than the standard reference-tracking NMPC in every configuration tested. The ensemble-style ϕ -uncertainty gate on the MPPI shield’s friction cone, evaluated under both the original tighten-cone formulation and a redesigned inflate-buffer formulation, degrades collision performance relative to a baseline that ignores live terrain updates entirely. The final system configuration therefore omits the ϕ -gate on the shield and uses the standard NMPC as the primary planner.

The joint (n, ϕ) identification of Section IV-A is established offline on the rich-excitation Latin-hypercube corpus; the closed-loop estimator deployed inside the NMPC currently regresses n only, and extending the live closed-loop path to

the joint head is a natural next step. The human-in-the-loop safety-filter comparison of Section VI-A is collected with a Logitech G29 steering rig under symmetric command/camera latency; the autonomous sweeps reported elsewhere serve as a controlled, repeatable complement that removes operator variance.

A direction for future work suggested by the negative result of Section IX-B is to design the estimator-to-shield interface as a low-pass mixture of the live estimate rather than as a direct overwrite. The hypothesised failure mode is that the shield’s safety calculations were calibrated against fixed terrain, and an unfiltered live update during evasive maneuvers introduces fast-time-scale non-stationarity that the safety cost is not designed to damp. A shield that attenuated the estimator update during high-cost epochs would test this hypothesis directly.

REPRODUCIBILITY

The complete sweep, including every figure and table in this paper, is reproduced by the following single command in the project’s `sim` conda environment:

```
python paper_scripts/run_paper_suite.py
-tier pilot
```

The exact result folders backing each figure are catalogued in `my_paper/paper_figures/README.md`.

ACKNOWLEDGEMENTS

This work builds on the Chrono::HIL framework [9] and Project Chrono [7], [5]. The 5G traffic profile is generated using the public N-HiTS-5G implementation released with Choi et al. [2].

REFERENCES

- [1] M. G. Bekker, *Introduction to Terrain-Vehicle Systems*. Ann Arbor, MI: University of Michigan Press, 1969.
- [2] Y.-H. Choi et al., “ML-based 5G traffic generation for practical simulations using open datasets,” *IEEE Commun. Mag.*, vol. 61, pp. 130–136, 2023.
- [3] J. Dallas et al., “Terrain adaptive trajectory planning and tracking on deformable terrains,” *IEEE Trans. Veh. Technol.*, vol. 70, no. 11, pp. 11255–11268, 2021.
- [4] R. Krenn and G. Hirzinger, “SCM – a soil contact model for multi-body system simulations,” in *Proc. Int. Soc. for Terrain-Vehicle Systems (ISTVS)*, Bremen, Germany, 2009.
- [5] R. Serban, M. Taylor, D. Negrut, and A. Tasora, “Chrono::Vehicle: template-based ground vehicle modelling and simulation,” *Int. J. Veh. Perf.*, vol. 5, no. 1, pp. 18–39, 2019.
- [6] H. Shibly, K. Iagnemma, and S. Dubowsky, “An equivalent soil mechanics formulation for rigid wheels in deformable terrain,” *J. Terramech.*, vol. 42, no. 1, pp. 1–13, 2005.
- [7] A. Tasora et al., “Chrono: an open source multi-physics dynamics engine,” in *High Performance Computing in Science and Engineering*, pp. 19–49, Springer, 2016.
- [8] H. Zhang et al., “Shared control of teleoperated vehicles with delay-compensated safety filtering,” in *Proc. IEEE Conf. Control Technol. Appl. (CCTA)*, pp. 192–199, 2025.
- [9] Z. Zhou et al., “A Chrono-based framework for large-scale traffic simulation with human-in-the-loop,” 2023.